



Saint Columban College  
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Pagadian City



## Scan Surprise

Scan Surprise 



Easy Forensics picoCTF 2024 shell browser\_webshell\_solvable qr\_code

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### Description

I've gotten bored of handing out flags as text. Wouldn't it be cool if they were an image instead?

You can download the challenge files here:

- [challenge.zip](#)

Additional details will be available after launching your challenge instance.

This challenge launches an instance on demand.

Its current status is: **NOT\_RUNNING**

[Launch Instance](#)

Hints 

1 2 3

67,519 users solved

 85% Liked 

 picoCTF{FLAG}

[Submit Flag](#)

**Author:** The Analyst: Hyposelenia

**Challenge:** Scan Surprise

**Category:** Forensics

**Date:** 04/02/26



## I. Objective

The objective of this challenge is to analyze a compressed file, identify its contents, and interpret visual data (QR code) to **retrieve hidden information**.

The goal is to extract the file, recognize the QR code, and scan it to **obtain the flag**.

## II. Background

In digital forensics, files may contain hidden or encoded data in different formats.

One such format is a **QR code**, which stores data in an image form. Instead of plain text, information is encoded visually and must be scanned to be understood.

In this challenge:

- The data was hidden inside a **ZIP file**
- The extracted file was a **QR code image**
- The QR code contained the **flag**

## III. Tool Used

- **Kali Linux (Oracle VirtualBox)** - Used as the working environment (optional but useful for practice)
- **unzip** - Used to extract the compressed file
- **xdg-open** - Used to open the image file
- **Online QR Code Scanner** - Used to decode the QR code

## IV. Methodology

**Note: this is solvable without Linux and Launching Instance**

1. Launched **Kali Linux** via Oracle VirtualBox
2. Downloaded the file using:  
*wget*



```
(kali㉿kali)-[~/Downloads]
$ wget https://artifacts.picoctf.net/c_atlas/14/challenge.zip
--2026-04-02 01:51:04-- https://artifacts.picoctf.net/c_atlas/14/challenge.zip
Resolving artifacts.picoctf.net (artifacts.picoctf.net) ... 18.172.21.12, 18.172.21.12
Connecting to artifacts.picoctf.net (artifacts.picoctf.net)|18.172.21.12|:443
HTTP request sent, awaiting response ... 200 OK
Length: 742 [application/octet-stream]
Saving to: 'challenge.zip'

challenge.zip
2026-04-02 01:51:05 (14.3 MB/s) - 'challenge.zip' saved [742/742]
```

3. Observed that the file was a **.zip** archive

```
(kali㉿kali)-[~/Downloads]
$ ls
challenge.zip
```

4. Extracted the contents:

***unzip <filename>.zip***

```
(kali㉿kali)-[~/Downloads]
$ unzip challenge.zip
Archive: challenge.zip
  creating: home/ctf-player/drop-in/
  extracting: home/ctf-player/drop-in/flag.png
```

5. Navigated through directories:

***cd Home***

```
$ ls
challenge.zip  home
(kali㉿kali)-[~/Downloads]
$ cd home
```

***cd ctf-player***

```
$ ls
ctf-player
(kali㉿kali)-[~/Downloads]
$ cd ctf-player
```

***cd drop-in***



```
$ ls
drop-in
(kali㉿kali)-[~
$ cd drop-in
```

6. Opened the image file:

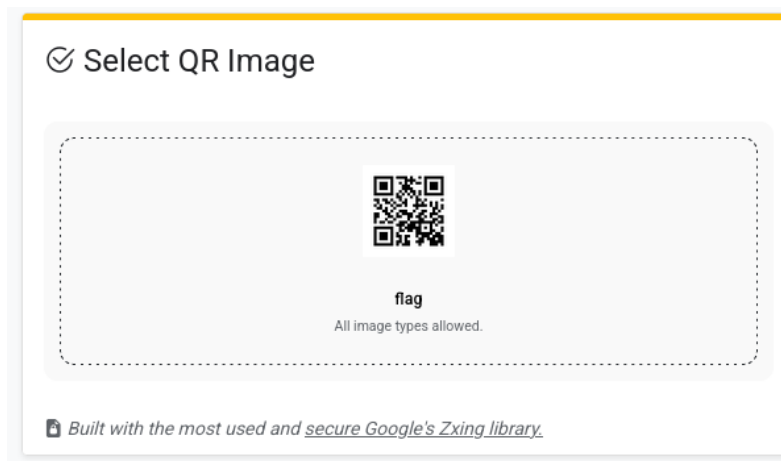
*`xdg-open <filename>.png`*

```
$ ls
flag.png
(kali㉿kali)-[~/Dow
$ xdg-open flag.png
```

7. Observed that the image is a **QR code**



8. Uploaded the image to an online QR code scanner



9. Retrieved the **flag**



## V. Result

**picoCTF{p33k\_@\_b00\_0194a007}**

 Scanned Data

```
picoCTF{p33k_@_b00_0194a007}
```

 Copy Results

## VI. Explanation

### What is a ZIP file?

A ZIP file is:

A container that holds files together

Like a **box** with items inside

You must **open (unzip)** it to see the contents.

### What does unzip do?

unzip file.zip

It means:

“Open the box and take out what’s inside”

### What does cd do?

**cd foldername**

Means:

“Go inside this folder”

### Analogy:

- Folders = rooms



- cd = walking into a room

### **What is xdg-open?**

xdg-open image.png

It means:

“Open this file using the default app”

### **Example:**

- Image → opens in image viewer
- PDF → opens in PDF reader

Think of it like:

Double-clicking a file on Windows

### **What is a QR Code?**

A QR code is:

A special image that stores information

### **Example uses:**

- links
- contact info
- payments

Instead of typing data, you just **scan it**

### **How does a QR code work?**

It is made of:

- black and white squares
- arranged in a pattern

Each pattern represents:

Binary data (0s and 1s)

### **Think of it like:**

- Secret message hidden in a picture



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- Scanner reads it → reveals the message

## VII. Conclusion

This challenge demonstrates how information can be hidden in visual formats such as QR codes.

By extracting the file, identifying the QR code, and decoding it using a scanner, the flag was successfully retrieved.

This reinforces a key concept in digital forensics:

***Data can be hidden in many forms, including images, and must be interpreted using the correct tools.***

— The Analyst: Hyposelenia